

BIOENERGETICS AND METABOLISM

FOCUS: ENERGY PRODUCTION

The metabolism is our body's way of producing energy. At the risk of stating the obvious, we need energy to move. Our ability to generate force and resist external stress is completely dependent on our ability to produce energy and to maintain energy production. If you can't run a marathon in under two hours, it's probably not due to a lack of endurance: it's almost certainly because you aren't able to produce the energy needed to run quickly enough and to maintain this level of energy for a whole two hours.

In order to create energy, the human body uses several different metabolic pathways: each with its own advantages, disadvantages and different characteristics. Each pathway draws energy from a substrate or fuel (phosphagens, carbohydrates and fats) to supply the energy needed for any given activity, movement or action. Understanding how energy is produced in each pathway and the consequences of these processes will give a better understanding of the reasons behind different training methods.

The energy-carrying molecule is known as ATP (adenosine triphosphate). When ATP reacts with a class of enzymes known as ATPases, one of its phosphate groups is separated off and the ATP is converted into adenosine diphosphate and inorganic phosphate, releasing a large amount of energy.

THE ATP-PCr SYSTEM

Also known as the phosphagen system or the anaerobic alactic system, this is the main source of energy for very short bursts of activity (10 to 15 seconds or less). Our muscles store small deposits of both ATP and creatine phosphate (PCr), which is another very high-energy molecule that actually produces more energy than ATP. Creatine phosphokinase helps release energy from PCr molecules by separating the creatine from the phosphate. This is a really fast way to produce energy and can be done without any oxygen (anaerobic). It also doesn't produce any lactate. However, these ATP and PCr stores are small and quickly depleted, so the body will switch to a different metabolic pathway if the demand for energy continues. As we'll see in more detail later on, climbing mainly involves intermittent contractions as we grip and release each hold, with just a short period of recovery between holds. As such, the ATP-PCr system supplies most of the energy needed in our forearms when climbing (Bertuzzi *et al.*, 2007).

ANAEROBIC GLYCOLYSIS

Glycolysis is the metabolic pathway that produces energy by breaking down glucose (*lysis* means 'separation' in Greek). Glucose makes up 99 per cent of the sugar in our bloodstream and is produced by digesting carbohydrates and by breaking down glycogen, which is stored in our liver and muscles. To simplify this, imagine glycogen as little bricks made up of several glucose molecules.

A total of 12 reactions are needed to produce energy from glucose, and no oxygen is required as this is an anaerobic pathway. The net return on this process is two molecules of ATP for every molecule of glucose, and the glucose is then broken down into pyruvic acid, which then becomes lactic acid. The lactic acid releases hydrogen ions (H⁺) which, when produced in large quantities, acidify the muscle fibres by lowering their pH. When this happens, the muscle fibres lose contractile capacity (Wilmore and Costill, 2007).

Therefore, although this system can produce high doses of energy, it also creates a build-up of waste products that end up inhibiting the production of energy. As such, it isn't an especially efficient system. Let's look at an analogy: imagine putting some wood (glucose) into a wood-burning stove to create heat (energy). When the fire has burnt out, you're left with ash (lactic acid). If you let the ash build up and don't clean it out, at some point the stove will become so full of ash (muscle acidification) that you won't be able to burn any more wood (muscle can't contract).

AEROBIC GLYCOLYSIS

Although glucose can be converted into energy anaerobically, this can also be done aerobically. In the presence of oxygen, glucose is still converted into pyruvic acid but this then becomes Acetyl-CoA, instead of lactic acid. Acetyl-

CoA is then used in another complex series of chemical reactions (the Krebs cycle) that produces two additional molecules of ATP (double the energy than in anaerobic glycolysis). In this cycle, the glucose is eventually converted into carbon and hydrogen. The carbon combines with oxygen to form carbon dioxide (CO_2), which is easily removed from the cell into the bloodstream and transported to the lungs to be exhaled. The remaining hydrogen (H^+) is used in another series of reactions (the electron transport chain) and ends up combining with oxygen to form water (H_2O), preventing the build-up of acid in our muscles and generating even more ATP. In total, 38 molecules of ATP can be produced from the oxidation of glucose (Wilmore and Costill, 2007). As such, this system is far more efficient and sustainable than the anaerobic production of energy.

FAT OXIDATION OR LIPOLYSIS

Although aerobic glycolysis sounds like a great way to produce energy, it 'only' produces between 1,200 and 2,000 kilocalories from our blood glucose and liver and muscle glycogen stores. This might be enough for a couple of goes on a boulder problem or just enough for a short, hard route, but it's nowhere near enough for a serious training session or a full day at the crag.

If 1 gram of fat stores 9 kilocalories of energy, a man of average weight, say 70 kilograms, who is in good shape and has a low body fat percentage (without being so low as to affect performance), say 12 per cent body fat, would have a total of 8.4 kilograms of fat, creating an energy reserve of 75,600 kilocalories. Enough energy to run three marathons, climb a bunch of routes and cycle back home ... Quite a difference, right?

It seems that burning fat can provide a much longer-lasting source of energy. However, this metabolic pathway is much more complex than the others discussed in this chapter, so all I'll say here is that the oxidation of fatty acids produces three times more ATP than the oxidation of glucose. What's more, the presence of oxygen means there is no build-up of lactic acid and no acidification of muscle fibres, meaning no loss of contractile capacity (Wilmore and Costill, 2007).



PHOTO: DARIO RODRIGUEZ

JAVI CANO AT THE CLIMB OPEN BOULDER FESTIVAL

THE ENERGY CONTINUUM

These energy-producing systems don't function independently: they are constantly interacting with each other. It would be illogical, inefficient and even dangerous to fully deplete one of these pathways before moving on to the next. Depending on the duration, intensity of effort, level of training and our individual hormones, one pathway will take precedence over the others but the interaction between them remains constant.

Climbing provides a good example of this interaction between pathways. As we usually need a lot of effort for a short period of time (when we're in contact with a hold), we're mainly using the ATP-PCr system. However, since our PCr (creatine phosphate) stores are practically emptied each time we latch a hold, the system uses the small window of time between holds to produce more energy via the oxidation of fats. This energy won't guarantee that we can keep climbing (we might not have enough power for those microscopic crimps, for example), but it will bind more phosphate to creatine, topping up our PCr stores and providing more high-intensity energy for the next hold.